AI SignBridge: Enhancing Sign Language Education and Real-Time Communication for Families and Friends of the Deaf

A Final Requirement on
COMP015- Fundamentals of Research

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Bachelor of Science in Information Technology

by

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Chapter 1

THE PROBLEM AND ITS BACKGROUND

Introduction

Organizations and government affairs acknowledge the essence of focusing on promoting inclusivity and equality towards the hearing-impaired community, especially since communication has become their greatest barrier between the hearing individual. Communication plays a vital role in society, as it shapes the individual socially, emotionally, mentally, and spiritually. Globally, there are 70 million with hearing disabilities; approximately 540,000 are reportedly using FSL in the Philippines. The World Federation of the Deaf states that this specific group of the deaf population is significantly more vulnerable and at risk of being 'unattended,' marginalized, and excluded from society, including deaf communities, when their social, linguistic, and communication needs remain unmet.

To maximize the potential of real-time communication, it must be supported by the integration of assistive technologies to promote inclusivity. The innovation of AI SignBridge is the key to addressing it correctly. By combining artificial intelligence and computer vision, we integrate real-time communication, and so the problems experienced by the deaf community will be solved.

The Persons with Disability Affairs Office in Sto. Tomas City provides assistance and services, including awareness campaigns and advocacy programs. In spite of providing these services, there was one time when San Augustin Church requested a sign language instructor in the Municipality of Sto. Tomas City, but they failed to provide due to the reason that there were no available interpreters at that moment. They didn't see that one coming, so there is no interpreter that can attend within minutes.

During the Government Affairs' years of operation, seminars and hearing aid tools are provided for the deaf and their families and friends. There are check-ups from different organizations voluntarily specifically for the deaf. The PDAO-Sto. Tomas City is always the number one bridge for communicating with other sponsors and for all the PWDs. The office makes sure that there is fair and enough assistance for those who are in need, especially to the deaf community.

At present, the mission of the PDAO-Sto. Tomas, to provide assistance, is still there, but as the years passed, many challenges occurred, such as the main problem that there is no consistent program for learning sign language. This main problem can lead to the deaf's well-being, relationships with their loved ones, and, more importantly, educational growth. The hearing-impaired person may continue to experience being 'unattended,' marginalized, and excluded from society as stated above by the World Federation of the Deaf. It can cause emotional distress not just to the deaf but to their loved ones as well now that not everyone can afford a private interpreter for them. Add that there is no financial assistance to the deaf community due to the reason that the municipality is more focused on the autism community since they are far more numerous than the deaf community in Sto. Tomas City.

It is for this reason why the proponents, being compassionate in thinking of a solution to world life problems, came up with the development of the AI-based system called "AI SignBridge." It will translate Filipino Sign Language in real time to text/speech and vice versa, have learning modules for the learners, and have a space where the deaf can share their own knowledge. The mobile-based system will bring convenience to the deaf people and to their families and friends. At long last, the proponents are developing a system that is available anytime, affordable, and accessible.

The AI SignBridge, an AI-powered application will be designed to empower sign language education and real-time communication between deaf individuals and hearing people especially with their families and friends, then it will utilize Machine Learning, Convolutional Neural Networks (CNN), and Generative Adversarial

Networks (GAN) to recognize and translate sign language gestures into text/speech and vice versa. Also, the application will serve as an interactive learning platform that will allow users–especially to families and friends of the deaf–to learn sign language and will function as an educational tool and communication assistant. Through this application, the deaf individuals will gain better understanding, will build stronger relationships, and will promote inclusive communication to their families and friends.

The development of AI SignBridge will aim to bridge the communication gap between the deaf and hearing people. This application will not only help deaf community to be heard and understood but also will serve as an educational tool especially for families and friends of the deaf individuals. This application with real-time translation and interactive learning tool will essentially foster mutual understanding, awareness, and appreciation of sign language as a medium communication between the deaf individuals and hearing people. The AI SignBridge will contribute to a more inclusive environment where every deaf individual will be able to interact with the hearing people without barriers, will be able to express themselves, and will participate equally in society.

The project will be conducted in order to assist the Persons with Disability Affairs Office (PDAO) of Sto. Tomas City in improving communication and understanding between the deaf community and the hearing population by developing a mobile-based assistive and educational application called "AI SignBridge."

AI SignBridge, an assistive and educational system, will have the following capabilities:

1. translate Filipino Sign Language gestures into text and spoken words and convert spoken words or text into sign-language visuals to enable smooth, real-time communication between deaf and hearing users.
2. provide interactive learning modules and exercises that will help families, friends, and interested individuals learn Filipino Sign Language

at their own pace, thereby encouraging better understanding and communication.

3. offer a space for deaf users to share knowledge and experiences, allowing them to contribute to awareness and inclusion within the community.
4. ensure accessibility and convenience, since the application will be mobile-based and can be used anytime and anywhere, making communication and learning more flexible and affordable.

Communication will continue to play a vital role in building relationships and understanding among people. However, many deaf individuals in Sto. Tomas City will continue to experience communication barriers due to the absence of consistent sign-language programs and the shortage of interpreters. This situation may lead to misunderstandings, social isolation, and limited access to education and opportunities. The purpose of the project will be to bridge this communication gap through the development of AI SignBridge, an assistive and educational mobile application that will promote inclusivity between the deaf and hearing communities. Once implemented, the system will provide a practical, accessible, and affordable solution that will strengthen family relationships, improve learning, and promote equal participation in the community.

Once the proposed AI SignBridge, which will promote inclusivity through the use of assistive technology, is implemented within Sto. Tomas City and possibly other localities, it will provide the following social and institutional significance particularly to the following:

Persons with Disability Affairs Office (PDAO – Sto. Tomas City).

- The PDAO will be able to extend more efficient services to the deaf community by providing an accessible and affordable communication platform. It will also help the office in its mission to promote inclusivity, awareness, and equality among persons with disabilities.

Deaf Community.

- Deaf individuals will greatly benefit from the application as it will enable them to communicate instantly with hearing people without the need for an interpreter. It will promote confidence, independence, and equal participation in daily activities.

Families and Friends of the Deaf.

- The interactive learning feature of AI SignBridge will allow family members and friends to study Filipino Sign Language at their own pace, strengthening emotional bonds and enhancing understanding within the family.

Educational Institutions.

- Schools will be able to use AI SignBridge as a supplementary learning tool for special-education and inclusivity programs, enabling teachers and students to develop basic sign-language skills.

Local Government Units (LGUs).

- The system will serve as a model for other municipalities aiming to provide inclusive digital services and support programs for persons with disabilities.

Researchers and Future Developers.

- The project will serve as a reference for future studies and innovations in developing assistive technologies for communication and education.

In summary, the implementation of AI SignBridge will create a more inclusive and understanding community by bridging communication gaps between the deaf and hearing populations. It will promote equality, accessibility, and social connection, proving that technology can serve as a bridge for meaningful human interaction.

Objectives of the Study

The main objective of the project will be to assist deaf community to communicate effectively, promote inclusivity and expose to opportunities similar to the hearing community. Specifically, it aims to:

1. design and develop an AI-based system called AI SignBridge which will be capable of:
    a. translating Filipino Sign Language (FSL) to text and audio, and vice versa.
    b. saving and loading through video the Filipino Sign Language (FSL) gestures.
2. evaluate the AI SignBridge in order to determine if it will technically and ethically comply to:
    a. ISO/IEC 25010 for software quality
    b. WCAG 2.1 for accessibility
    c. RA 10173 for data privacy
    d. RA 11106 for Filipino Sign Language Act
3. prepare a plan for the deployment of the AI SignBridge.

Scope and Limitation of the Study

The study will focus on developing and evaluating an AI-powered mobile application called AI SignBridge designed to enhance sign language education and real-time communication between the deaf and hearing communities. The application will specifically serve the deaf communities in the Persons with Disability Affairs Office (PDAO) of Sto. Tomas City.

The system's application will integrate Machine Learning, Convolutional Neural Networks, and Generative Adversarial Networks to identify Filipino Sign Language (FSL) gestures and will convert them into text and speech, and vice versa. Additionally, the system will provide interactive learning modules that will allow deafs, their families and friends, as well as interested hearing users, to learn sign language at own pace. The study will center on the design, development, and testing of the application's major features, which include real-time FSL-to-text and speech translation, interactive learning modules, offline accessibility for saved lessons, and mobile usability for convenience and portability.

The system will be limited to translating only static and dynamic FSL gestures included in the trained dataset and will only provide basic FSL lessons, excluding advanced signs or complex sentence structures. This limitation is caused by practical constraints such as the scope available annotated in FSL datasets and resources rather than technical restrictions in the selected AI architecture or software platform. However, the technical design includes the use of CNNs, LSTMs, and scalable frameworks and is beyond the basic signs to include advanced gestures and continuous signing capabilities.

The system will operate on Android and iOS devices during the development phase and support Filipino and English languages in text-to-speech conversion. Only lessons and resources that have been saved will be accessible offline; real-time AI translation will still need an active internet connection. Additionally, evaluation will focus on system correctness, usability, accessibility, performance, and user learning progress—and will not address long-term behavioral consequences.

The application is not intended for non-literate users, as written text remains essential for instruction and communication. Target users include deaf individuals registered under PDAO-Sto. Tomas City, their families, friends, and assigned PDAO personnel.

The study will be carried out within the academic year 2025 to 2026, aligning with standards such as ISO/IEC 25010, WCAG 2.1, RA 10173, and RA 11106.

Purpose and Description

This study will contribute to bridging the gap between the deaf and the hearing community. We, researchers, will aim to provide a technological innovation that will combine artificial intelligence and computer vision to translate Filipino Sign Language (FSL) into text-to-speech, and vice versa, in real time. Through this, the study will become vital in addressing one of the most pressing challenges faced by the deaf—the lack of accessible, efficient, and affordable means of communication.

For the Deaf Community, the system will provide an accessible tool that will allow them to interact with others without always relying on an interpreter. It will support their confidence, independence and equal participation in every social environment.

For the Families and Friends of the Deaf, the system will serve as an accessible and interactive learning tool for understanding and practicing sign language. It will strengthen emotional bonds and will enhance mutual understanding, reducing the feeling of isolation experienced by many deaf individuals and their loved ones.

For the Persons with Disability Affairs Office (PDAO) - Sto. Tomas, the system will serve as an assistive and educational tool that will support the agency's mission to provide fair and sufficient services to persons with disabilities. Also, it will offer a sustainable way to continue sign language learning programs even in the absence of available instructors.

For Educators and Researchers, AI Signbridge will serve as a reference and foundation for future studies and innovations in the field of assistive technology, artificial intelligence, and inclusive education. It will contribute new insights into the application of machine learning, convolutional neural networks (CNN), and generative adversarial networks (GAN) for real-time language translation.

For the Field of Information Technology, this project will exemplify how technology can be utilized to address real-world social issues, highlighting the ethical and technical integration of AI to serve marginalized communities.

In essence, the study will be significant as it will not merely introduce a technical innovation but also will embody a social advocacy—using technology to bridge communication barriers, promote inclusivity, and empower the deaf community.

Theoretical Framework

The AI SignBridge system will draw on four key theoretical foundations that will guide its design and use. First, the Extended Unified Theory of Acceptance and Use of Technology (UTAUT2) will highlight factors such as usefulness, ease of use, social influence, and enjoyment, showing that features like real-time Filipino Sign Language (FSL) translation, interactive learning modules, and offline access will encourage adoption by deaf users, their families, and hearing learners (Kim et al., 2024). Second, its technical backbone will rely on Artificial Intelligence and Neural Network principles, with machine learning models such as Convolutional Neural Networks (CNNs) and Generative Adversarial Networks (GANs) enabling accurate recognition of static and dynamic FSL gestures and their conversion into text or speech for real-time communication (El Samad, Nasserddine, & Kheir, 2023). Third, Human-Computer Interaction (HCI) and User-Centred Design (UCD) will guide the system's interface to ensure it will be intuitive, accessible, and inclusive (Kurosu, 2021). Finally, Landmark-Based Feature Extraction using MediaPipe will translate visual information from the hands, face, and body into structured coordinates, allowing AI SignBridge to model gesture sequences with temporal networks like GRUs and to provide accurate, real-time translation and feedback (Subramanian et al., 2022). Together, these foundations will create a system that will be not only technically robust but also user-friendly, promoting sign language learning, fostering communication between deaf and hearing individuals, and supporting adoption across diverse users.

Conceptual Framework

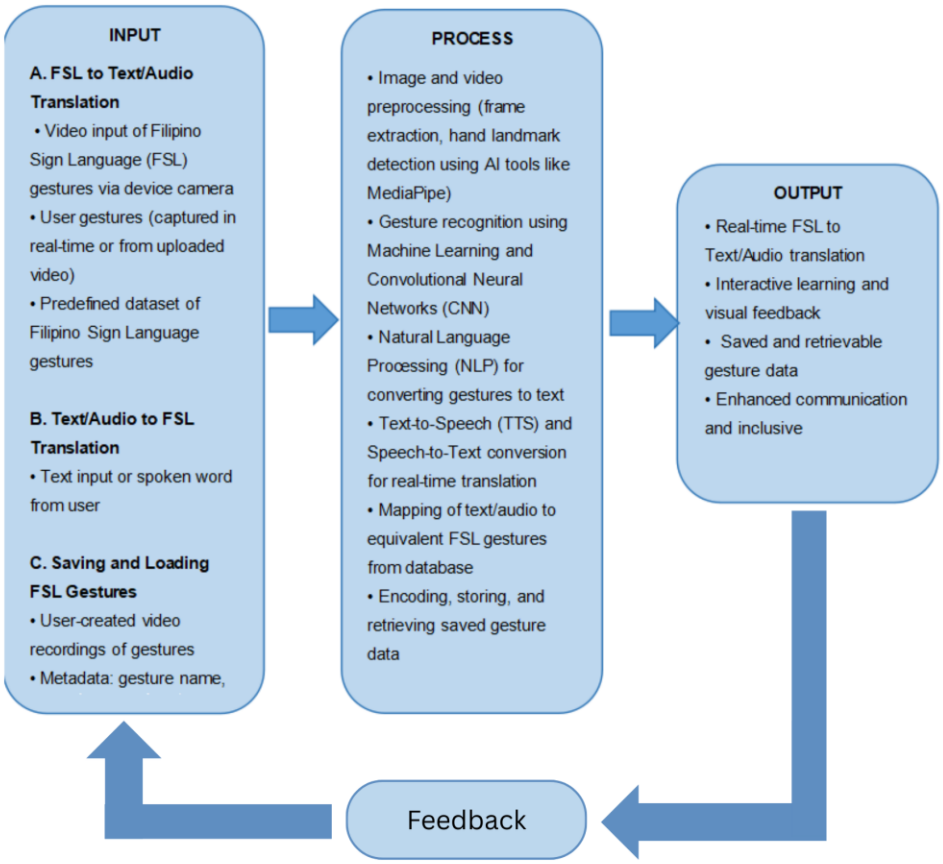

Fig. 1: IPO Chart

Chapter 2

REVIEW OF RELATED LITERATURE, STUDIES, AND SYSTEMS

Sign Language Recognition and Gesture-Based AI Models

Research consistently emphasizes the feasibility of vision-based approaches for sign language recognition, particularly in the Filipino context. Co et al. (2024) demonstrated that MediaPipe hand landmarks combined with LSTM models can identify Filipino Sign Language (FSL) gestures efficiently on low-resource devices, making them suitable for mobile phones and embedded systems. Rastgoo et al. (2021) reinforced this by highlighting CNN, RNN, and LSTM architectures as industry standards for gesture recognition without wearable sensors. Similarly, Pilare et al. (2024) showed that basic FSL words can be recognized using MediaPipe and LSTM, though dataset limitations remain. Complementary findings by Rahman et al. (2019), Suardi et al. (2021), and Saiful et al. (2022) confirm CNN's effectiveness in processing visual gestures, achieving accuracies above 98%. Local studies by Montefalco et al. (2021; 2023) further validated deep learning models such as LSTM and ResNet for FSL recognition, emphasizing the importance of facial expressions and non-manual features. Together, these works validate the scalability and accuracy of AI-powered, camera-based recognition systems for inclusive communication, while also pointing to the need for larger datasets and real-world testing.

Machine Learning and Deep Learning for Real-Time Translation

Advancements in machine learning highlight the importance of hybrid architectures for capturing both spatial and temporal gesture data. Kumari and Anand (2024) demonstrated that CNN-LSTM hybrids with attention mechanisms improve recognition accuracy, while Cayme et al. (2024) achieved up to 99% precision and recall in real-time FSL recognition using CNN-LSTM models. Barve et al. (2021), Zhang and Jiang (2024), and Alam et al. (2024) reinforced the role of CNNs, RNNs, and transformer-based models in overcoming challenges such as signer variability, lighting conditions, and environmental noise. Alam et al. also emphasized the potential of mobile-based assistive technologies despite computational limitations. Studies like Murillo et al. (2021) and Cabigting et al. (2022) applied these models in real-time systems, achieving high accuracy in bidirectional FSL-to-text/speech translation. International works such as Adithyaraaj et al. (2025) demonstrated transformer-based NLP models integrated with CNNs for gesture recognition, achieving real-time interpretation of hand gestures and facial expressions. Collectively, these findings underscore the need for robust, scalable, and mobile-friendly AI architectures to enable real-time communication between deaf and hearing individuals.

Assistive Technologies and Inclusive Communication Tools

Assistive technologies play a pivotal role in bridging communication gaps between deaf and hearing individuals. Rodriguez et al. (2023) and Rashid et al. (2024) emphasized mobile applications and AR-based tools as vital for daily communication, while Pan et al. (2025) and Putri et al. (2025) highlighted user-centered HCI designs that involve Deaf and Hard-of-Hearing (DHH) individuals directly in development, improving inclusivity and engagement. Local studies such as Abalos et al. (2024) and Naraja et al. (2025) extended this by developing mobile applications like SignTalk, showcasing GANs and CNNs for real-time translation. Pilare et al. (2024) also demonstrated high accuracy in determining FSL hand gestures using MediaPipe and LSTM, though gesture coverage remains limited. These findings collectively support the development of AI SignBridge as a mobile-based assistive tool that is both inclusive and practical, ensuring accessibility and community-driven participation.

Real-Time Communication Systems and Speech-to-Sign Interfaces

Papatsimouli et al. (2022) and Mohammedali et al. (2022) reviewed real-time sign language translation systems, showing how gesture recognition can convert hand and facial expressions into text or speech. Kaur et al. (2024) achieved over 97% accuracy using InceptionResNetV2 for ASL-to-speech conversion, while Shashidara et al. (2025) and Pokanati et al. (2025) explored bidirectional AI interfaces using CNNs and LSTMs for speech-to-sign and sign-to-text systems. Tikore (2025) reinforced the role of MediaPipe and TensorFlow in achieving high real-time accuracy. Local studies such as Cabigting et al. (2022) demonstrated two-way communication using Jetson Nano, MediaPipe, and Google's TTS/STT services, achieving accuracy above 93%. These systems demonstrate the potential of AI-powered translation tools to enable barrier-free communication and justify the integration of real-time modules in AI SignBridge.

Accessibility and Inclusive Education Technologies

Studies highlight the importance of inclusive design in educational technologies for DHH learners. Nasabeh and Meliá (2024) proposed IoT-based ecosystems for emotional and social support, while de Godoi et al. (2020) and Rodríguez-Correa et al. (2023) noted the need for user-centered, deployable systems beyond prototypes. Flórez et al. (2017) and Metatla et al. (2018) emphasized interactive storytelling and universal design principles to ensure equal access, while López et al. (2025) and Chen et al. (2025) advocated for technologies that address emotional awareness through Automatic Emotion Recognition (AER). Fernandez et al. (2025) and Velasco et al. (2024) demonstrated mobile applications with gamification and gesture recognition to enhance FSL learning. Holt (2019) added that wearable technologies like hearing aids and cochlear implants, while helpful, still leave gaps in educational inclusion. Together, these studies highlight how inclusive, gamified, and emotionally aware technologies can strengthen education and social participation — principles embedded in AI SignBridge's interactive learning modules.

AI-Based Language Translation and Communication Aids

Cayme et al. (2024), Camgoz et al. (2020), and Yin and Read (2020) demonstrated the efficiency of CNN-LSTM and transformer-based models in translating sign language into text and speech. Systems like Text2Sign (Stoll et al., 2020) and mobile learning tools by Banag (2023) and Lawana (2024) showcased gamified, user-centered designs that enhance accessibility and engagement. Ahmad Faudzi et al. (2023) emphasized culturally appropriate interfaces for successful adoption. International studies such as Cate (2025) extended AI applications to crisis communication, integrating IoT sensors and cloud analytics for emergency alerts. Local works by Montefalco et al. (2021; 2023) and Battina & Lakshmisri (2022) highlighted both the promise and limitations of AI-driven communication aids, pointing to the need for broader gesture sets and more organic, camera-based alternatives. These findings collectively reinforce the relevance of AI SignBridge as both a translation tool and a community-driven platform for inclusive communication.

Technical Background

AI SignBridge will utilize Convolutional Neural Networks (CNNs) for gesture and image recognition to analyze real-time visual input and accurately match FSL gestures. Generative Adversarial Networks (GANs) will improve model reliability and increase dataset diversity for both static and dynamic sign recognition (Géron, 2019; Goodfellow et al., 2016). The system combines CNNs for spatial feature extraction with RNNs or LSTM networks for temporal modeling of continuous movements and gesture transitions (Howard et al., 2017). This hybrid CNN-LSTM approach ensures robust recognition of signing sequences through frame-by-frame processing (Pigou et al., 2015). The computer vision pipeline employs MediaPipe to extract real-time hand, face, and landmark features with 3D spatial data, achieving at least 30 frames per second through model quantization and pruning optimizations (Lugaresi et al., 2019; Bradski & Kaehler, 2008).

The application will be developed using Python for AI model training and either Flutter with Dart or React Native with JavaScript/TypeScript for cross-platform deployment. TensorFlow Lite and PyTorch Mobile will enable efficient on-device inference, while SQLite or native file systems will support offline functionality (Tan & Le, 2019). The user interface will follow WCAG 2.1 standards with accessible navigation, touch-optimized controls, and screen reader support. Text-to-speech and speech-to-text functionality will handle both Filipino and English to enhance inclusivity (Wang et al., 2017).

AI SignBridge will implement AES-256 encryption for data at rest and TLS 1.3 for data in transit, using secure storage solutions like SQLCipher. The system follows "privacy by design" principles with user consent, data minimization, and compliance with the Philippine Data Privacy Act (Republic Act No. 10173). A layered architecture with separated modules for presentation, application, machine learning, and data management will ensure maintainability and scalability, supporting efficient processing from input to output in both online and offline modes (Bazarevsky et al., 2020).

Synthesis

Despite numerous advancements in sign language recognition systems, most existing technologies remain limited in scope, focusing on foreign sign languages, requiring specialized hardware, or lacking real-world deployment. In the Philippines, Filipino Sign Language (FSL) is underrepresented in both research and assistive technology, leaving a critical gap in inclusive communication tools for the Deaf community.

The present study addresses this gap by developing AI SignBridge, a mobile-based assistive and educational system that integrates real-time FSL translation with interactive learning modules for families and friends of the Deaf. Unlike prior works that focus solely on gesture recognition or translation, AI SignBridge combines machine learning, CNNs, GANs, and MediaPipe landmark extraction to deliver a culturally relevant, user-centered, and deployable solution.

Literature reviewed supports the feasibility and necessity of such systems, but also highlights persistent challenges—such as signer variability, dataset limitations, and lack of mobile optimization—which this project directly tackles. Moreover, AI SignBridge aligns with Philippine laws (RA 11106, RA 10173), accessibility standards (WCAG 2.1), and software quality benchmarks (ISO/IEC 25010), ensuring ethical and sustainable implementation.

Therefore, this study is not a duplication of existing works but a novel contribution that empowers the Deaf community, strengthens family relationships, and promotes inclusive education through a localized, technically robust, and socially impactful solution.

Chapter 3
METHODOLOGY

This chapter will describe the research design and methods that will be used to evaluate the AI SignBridge application. It will cover the participants, sampling techniques, research instruments, and data collection procedures. The methodology is designed to gather data to assess the system's functionality, usability, performance, accessibility, and overall user satisfaction.

Research Design

This study will employ a Developmental–Descriptive Research Design to systematically design, develop, and evaluate AI SignBridge, a mobile-based assistive and educational application intended to enhance sign language communication between deaf individuals and hearing users such as families, friends, and PDAO personnel. The design will integrate both system development processes and descriptive evaluation methods to ensure that the application will meet user needs and adhere to recognized standards for software quality and accessibility.

Software Development Methodology

2.1 Development Methodology

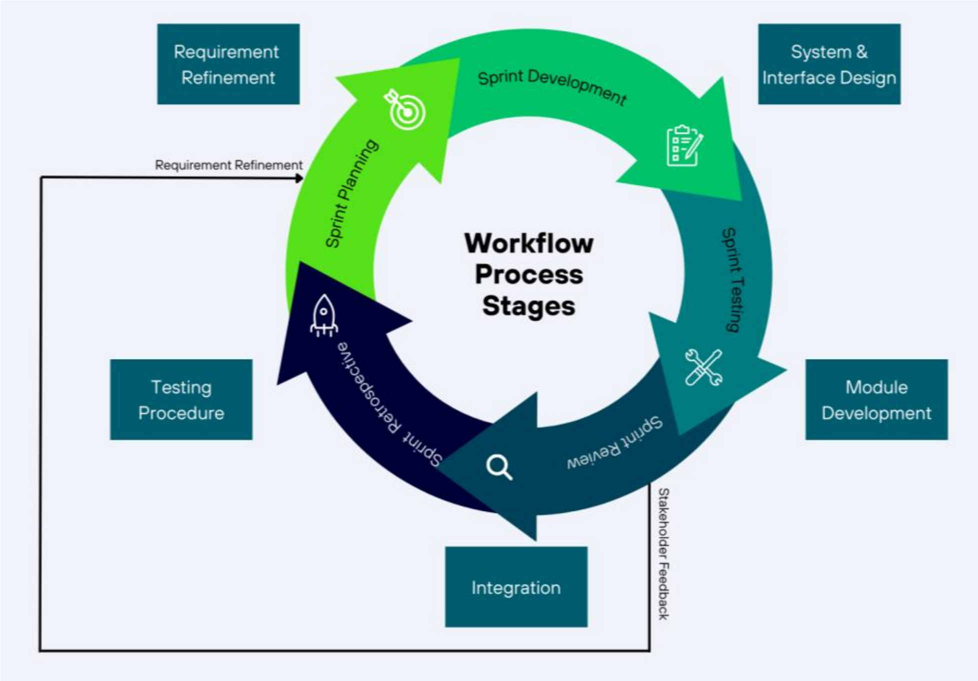

Fig. 1: Agile Development Process

The development of AI SignBridge will adopt the agile methodology followed by an iterative and incremental development approach. Agile will be used to require continuous user feedback, model improvements, and flexibility to adjust the changes in design and features, especially to the system involved in developing sign language recognition using machine learning. Through this agile method, the development team will deliver functional requirements at the end of each sprint, allowing users—including families and friends of the deaf individuals—to test the early versions of the system and provide feedback for requirements needed.

2.2 Agile Sprint Structure

The whole development process will be divided into several 1–2 weeks sprints, each producing a working module of the AI SignBridge system.

1. **Sprint Planning**

   During sprint planning, the team will identify and prioritize user stories such as:

   - As a deaf user, I want my Filipino Sign Language gestures to be translated into text or spoken words so that I can communicate more effectively with hearing individuals.
   - As a hearing user, I want my spoken words or text to be converted into sign-language visuals so that I can interact smoothly with deaf individuals even without prior knowledge of FSL.
   - As a PDAO personnel, I want to oversee communication sessions and ensure translation accuracy so that the application can be reliably used in official programs and inclusive initiatives.
   - As a community member, I want a space to share experiences and insights so that awareness and inclusion within the deaf community are promoted.
   - As a mobile user, I want the application to be accessible anytime and anywhere so that communication between deaf and hearing individuals remains flexible and convenient.

2. **Sprint Development**

   Each sprint will involve the following activities:

   - Developing UI components such as the camera interface for gesture input, translation display screens for text and speech output, and communication panels that facilitate real-time interaction between deaf and hearing users.
   - Implementing machine learning features including gesture detection and model prediction to ensure accurate recognition of Filipino Sign Language gestures and their conversion into text or spoken words.

- Integrating the backend, frontend, and database systems to support secure user authentication, manage translation logs, and maintain smooth data flow between the recognition module, user interface, and storage components.

3. **Sprint Testing**

   After development, the team will perform:
   - Unit testing (for each feature)
   - Integration tests (UI, ML model, database)
   - Fixing bugs found during testing

4. **Sprint Review**

   At the end of each sprint, the working module of AI SignBridge will be presented to key stakeholders for evaluation. These stakeholders include:
   - Members of the deaf community, who will test the accuracy and usability of the real-time Filipino Sign Language (FSL) translation features.
   - Family and friend users, who will assess how effectively the system supports communication with deaf individuals through text-to-FSL and FSL-to-speech translation.
   - PDAO personnel, who will evaluate the application's relevance to inclusive communication initiatives and its potential integration into community programs.

Feedback gathered during the sprint review will be used to refine requirements, improve translation accuracy, and enhance the overall user experience in subsequent iterations.

5. **Sprint Retrospective**

Following each sprint, the development team will conduct a retrospective session to critically evaluate the process and outcomes of the iteration. This activity will involve analyzing what aspects of the sprint were successful, identifying challenges encountered during development and testing, and determining strategies for improvement in subsequent cycles. The retrospective will serve as a structured reflection point, allowing the team to refine workflows, enhance collaboration, and address technical, or usability issues observed during stakeholder reviews. By systematically documenting lessons learned, the team ensures that each iteration contributes to the continuous improvement of AI SignBridge, ultimately leading to a more reliable, accessible, and user-centered application.

2.3 Development Stages

Although Agile development is iterative, the following stages will guide the project cycle to ensure coherence and quality:

- **Requirement Refinement**
    - Continuously review and refine system requirements based on stakeholder feedback and prototype testing.
    - Focus on functional requirements such as:
        - Gesture-to-text translation
        - Text-to-FSL visualization
        - Speech-to-FSL conversion
    - Ensure alignment with the communication needs of deaf individuals, families, and PDAO personnel.

- **System and Interface Design**
    - Prepare evolving design deliverables, including:
        - User interface layouts for the camera module, translation display screens, and communication panels
        - System architecture diagrams showing integration of machine learning models for gesture recognition

        - Entity-Relationship Diagrams (ERDs) for user authentication, translation logs, and community interactions
        - Data flow diagrams mapping gesture input → model detection → system response

- **Module Development**
    - Incrementally build system modules such as:
        - Real-time FSL recognition and translation
        - User authentication and secure profile management
        - Text-to-FSL and FSL-to-speech interfaces
        - Community-sharing space for deaf users to contribute experiences and insight

- **Integration**
    - After each sprint, integrate new modules into the working system.
    - Ensure compatibility and smooth data flow between:
        - Machine learning recognition engine
        - Frontend user interface
        - Backend database systems
    - Maintain system stability, scalability, and responsiveness to user needs.

2.4 Testing Procedures

The entire development cycle of **AI SignBridge** will consist of testing at multiple levels to ensure accuracy, stability, and usability:

- **Unit Testing**
    - Test individual machine learning functions such as gesture detection and frame processing.

- Validate UI components including buttons, forms, and translation display screens.
- Check backend functionalities such as API endpoints and secure handling of user data.

- **Integration Testing**
  - Verify that the machine learning model integrates correctly with the camera interface.
  - Ensure the UI displays translation predictions accurately.
  - Confirm database operations such as saving translation logs and user progress function properly.

- **System Testing**
  - Evaluate the platform's overall performance, including real-time detection accuracy.
  - Measure response time and system stability under different conditions.
  - Test compatibility across multiple devices (Android and iOS).
  - Assess user interface usability and accessibility compliance.

- **User Acceptance Testing (UAT)**
  - Conduct testing with members of the deaf community to validate translation accuracy and usability.
  - Involve families and friends to assess communication effectiveness in real scenarios.
  - Gather stakeholder feedback to confirm that the system meets communication and inclusivity needs.

- **Regression Testing**

- Perform testing whenever new features or machine learning improvements are introduced.
- Ensure that existing functionalities—especially gesture recognition and translation—remain stable and unaffected.

2.5 Testing Tools and Techniques

Testing will employ the following tools and techniques to ensure accuracy, usability, and reliability:

- **Manual testing** for user interface behavior and translation interactions.
- **Dataset evaluation** to measure the accuracy of machine learning models for gesture recognition.
- **Confusion matrices and performance metrics** to assess recognition precision and error rates.
- **Logs and error tracking tools** to identify, document, and resolve system issues.
- **Test scripts (optional)** for repeated validation of machine learning models and translation outputs.

2.6 Acceptance Criteria

A feature will be considered complete when the following conditions are met:

- **Code implementation and documentation** are finalized and properly maintained.
- **Testing requirements** are satisfied, including successful unit, integration, and sprint-level tests.
- **Machine learning models** achieve the minimum acceptable accuracy for gesture recognition and translation.
- **User interface** is functional, accessible, and user-friendly across devices.
- **Stakeholder approval** is obtained during the Sprint Review, confirming that the feature meets communication and usability needs.

Research Participants

The participants of this study will consist of **54 selected individuals**, including members of the deaf community registered under the Persons with Disability Affairs Office (PDAO) of Sto. Tomas City, along with their families and friends who will serve as primary communication partners. These individuals will be chosen through purposive sampling because they will represent the actual users who will benefit from real-time Filipino Sign Language (FSL) translation and the educational features of AI SignBridge. Participating as an administrative stakeholder are the PDAO personnel, who will oversee deaf-related programs and are responsible for employing inclusive communication initiatives inside the locality. Their combined perspectives will provide essential insights into the system usability, translation accuracy, and overall relevance of the application to the communication needs of the deaf and hearing communities.

Research Instruments

System Capabilities

	5	4	3	2	1
1. The system will accurately translate FSL gestures into readable text.					
2. The system will accurately translate FSL gestures into spoken audio.					
3. The system will correctly convert text/speech into animated or recorded FSL visuals.					
4. The system will allow users to save and load gesture videos easily.					
5. The system will provide useful and structured learning modules for FSL.					

Software Quality (ISO/IEC 25010)

	5	4	3	2	1
6. The system's core functions will behave as expected (Functional Suitability).					
7. The system will respond fast enough for real-time translation (Performance Efficiency).					
8. The system will operate reliably with minimal errors (Reliability).					
9. The system will be easy to learn and use (Usability).					
10. The system will be maintainable for future updates (Maintainability).					

Accessibility (WCAG 2.1)

	5	4	3	2	1
11. Text, icons, and visuals will be perceivable and legible (Perceivability).					
12. Navigation and control elements will be operable for all users (Operability).					
13. Instructions, messages, and help will be understandable (Understandability).					
14. The app will work across devices and assistive tools (Robustness).					

Data Privacy & Legal Compliance (RA 10173, RA 11106)

	5	4	3	2	1
15. The app will clearly inform users about how their data (including videos) will be used (Transparency).					
16. The app will protect user data through proper security measures (Data Protection).					
17. The system will respect the cultural and linguistic accuracy of Filipino Sign Language (FSL Act compliance).					

Deployment Readiness & Acceptance

	5	4	3	2	1
18. I will be willing to use or recommend AI SignBridge for everyday communication.					
19. The system will be appropriate for use in PDAO programs and community activities..					
20. Overall, the system will improve communication between deaf and hearing users.					

Weighted Arithmetic Mean

$$\bar{x} = \frac{\sum_{i=1}^{n} x_i}{n}$$

Where:

- $\bar{X}$ = Weighted Arithmetic Mean of the Likert-scale responses
- $\sum x_i$ = Sum of all numerical values assigned to the respondents' answers
- x_i = Numerical value of each respondent's Likert-scale rating
- **n** = Total number of respondents

In this study, the formula will be used to compute the mean score of each survey item measured using a 5-Point Likert Scale, where responses are quantified as follows:

Scale Description	**Numerical Value**
Strongly Agree	5
Agree	4
Neutral / Fair	3
Disagree	2
Strongly Disagree	1

Each respondent's answer will be assigned its corresponding numerical value. The total of all responses for a given item will be divided by the number of respondents to obtain the mean score, which represents the overall perception or evaluation of the system.

Interpretation Scale

Mean Range	Interpretation
4.50 – 5.00	Strongly Agree — Excellent / Very High
3.50 – 4.49	Agree — High
2.50 – 3.49	Neutral / Fair — Moderate
1.50 – 2.49	Disagree — Low
1.00 – 1.49	Strongly Disagree — Very Low

This method will allow the researchers to quantitatively analyze user perceptions regarding the functionality, usability, accessibility, performance, and overall effectiveness of AI SignBridge, ensuring alignment with the study's evaluation objectives and standards.

Decision rule:

- Domain mean ≥ 3.50 → acceptable / ready (satisfactory).
- Domain mean < 3.50 → needs improvement / review.

The study will utilize a structured survey questionnaire as the primary research instrument to evaluate the AI SignBridge application in terms of functionality, usability, performance, accessibility, and overall user satisfaction based on the

ISO/IEC 25010 and WCAG 2.1 standards. The questionnaire will employ a 5-point Likert scale to quantify user perceptions from deaf individuals, their families, and PDAO personnel. Additionally, to gather qualitative insights relating to user experiences, an interview guide will be utilized. It will also be used for communication challenges and suggestions for enhancing real-time Filipino Sign Language (FSL) translation and learning features. For documenting the user interaction behaviors, accuracy of gesture recognition, and accessibility performance, the observation checklists will be used during system demonstration sessions. Altogether, these instruments will provide comprehensive and essential data useful for evaluating the effectiveness and impact of AI SignBridge in enhancing communication between deaf and hearing users.

Data Collection Procedure

The data collection procedure will begin by securing approval from the Persons with Disability Affairs Office (PDAO) of Sto. Tomas City to conduct the study with deaf individuals, their families, and PDAO personnel. After the permission is given, the proponents will administer a structured survey questionnaire containing Likert-scale items that is aligned to ISO/IEC 25010 and WCAG 2.1 standards to evaluate the usability, accessibility, and translation accuracy of the AI SignBridge application. To gather more deeper insights, the proponents will further conduct interviews with all the selected interviewee. The interview will be conducted to determine the communication needs and expected system performance. In addition, observation checklists will be utilized during scheduled system demonstrations to document user interactions and assess gesture recognition accuracy. All data will be collected ethically, with informed consent obtained from participants, and the gathered responses will serve as the basis for analyzing the system's effectiveness and its alignment with the study's objectives.

Functional and Non-Functional Requirements

Requirement	Description	Relevance to Research Problem
User Authentication	Secure login and registration for deaf users, families, and PDAO staff.	Ensures privacy and secure access, protecting sensitive user profiles and gesture data.
User Roles and Permissions	Provides different access levels for deaf individuals, family members, and PDAO personnel.	Supports organized user management and ensures proper access to translation, learning, and administrative tools.
FSL Gesture Recognition	Detects and processes Filipino Sign Language gestures using CNN and AI models.	Directly addresses communication barriers by enabling real-time recognition of FSL.
Text-to-FSL and Speech-to-FSL Translation	Converts text or spoken words into animated or pre-recorded FSL gestures.	Enables hearing individuals to communicate effectively with the deaf, even without sign language knowledge.
FSL-to-Text and FSL-to-Speech Translation	Translates FSL gestures into readable text and spoken audio output	Facilitates instant communication for deaf users, reducing dependence on interpreters.
Interactive Learning Modules	Provides structured FSL lessons, exercises, and progress tracking.	Addresses the lack of sign-language education programs and empowers families to learn FSL at their own pace.
Gesture Saving and Loading	Allows users or administrators to save, store, and replay gesture videos.	Supports continuous learning, training, and dataset expansion for improved system accuracy.
Knowledge-Sharing Community Space	A section where deaf users can share experiences, videos, and insights.	Promotes inclusivity and empowers the deaf community by giving them a platform to contribute.
Notifications and Alerts	Sends real-time updates regarding saved gestures, learning progress, or system reminders.	Keeps users informed and engaged, improving learning continuity and communication.

Table. 1: Functional Requirements

Requirement	Description	Relevance to Research Problem
Data Security and Privacy Compliance	Ensures encryption, secure storage, and compliance with RA 10173 (Data Privacy Act).	Builds user trust and protects personal and biometric information of deaf individuals.
System Usability and Accessibility	Ensures the app's interface is intuitive, accessible, and compliant with WCAG 2.1.	Makes the system friendly for both deaf and hearing users, improving overall user experience.
System Performance and Accuracy	Ensures fast processing, accurate gesture recognition, and reliable translation.	Prevents miscommunication and ensures the system works effectively in real time.
Reliability and Maintainability	Provides stable operation, error handling, and easy updates for continuous improvement.	Ensures long-term sustainability and system readiness for community use.
Compatibility and Portability	Supports Android and iOS devices with scalable architecture.	Maximizes accessibility by allowing users to access the system on various devices.

Table. 2: Non-Functional Requirements

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